KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 14 · MATH

Cargo Day at PortMiami — Teacher Lesson Plan & Answer Key

Learning objectives: Students convert customary units of length (in, ft, yd), weight (oz, lb, T), and capacity (c, pt, qt, gal), and metric units (cm, m, km, g, kg, mL, L) using conversion tables and the rule "big to small, multiply; small to big, divide." Students convert hours and minutes, find elapsed time, regroup mixed units, and solve multi-step word problems, explain their reasoning, and find mistakes.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Andre, who spends the day at PortMiami with his Aunt Guerline (crane operator), Mr. Alvarez (refrigerated fruit containers), and Ms. Raman (cargo planner); student types first and last name so results are sent.
2. Slide 1 — Warm-Up: Multiplying Fractions Review (8 pts): fraction × whole number, fraction × fraction, scaling, and a fraction of a measurement (Week 13). If a student misses two or more, rework one on paper before moving on.
3. Slide 2 — Pencil Break (Pre): student completes the Pre-Lesson Warm-Up sheet off screen.
4. Slide 3 — Hook: PortMiami on Dodge Island in Biscayne Bay; containers are 20 or 40 feet long and 8 feet wide; refrigerated containers keep fruit cool. Andre's list previews every unit family.
5. Slide 4 — Picture It: a yardstick shows 1 yard = 3 feet = 36 inches. Teach the rule "big unit to small unit, multiply; small unit to big unit, divide" and the reasonableness check. Read the feet-to-inches conversion table together and name its rule (× 12).
6. Slide 5 — Practice: inches, feet, and yards, including a conversion table (12 pts).
7. Slide 6 — Teach: weight (1 lb = 16 oz, 1 T = 2,000 lb) and capacity (bar model: 1 gal = 4 qt = 8 pt = 16 c). Walk through 2 gallons to cups one step at a time.
8. Slide 7 — Practice: weight and capacity, both directions (12 pts).
9. Slide 8 — Teach: metric units and prefixes (kilo- = 1,000; centi- = one hundredth; milli- = one thousandth). Ask the student which units are bigger before converting.
10. Slide 9 — Practice: metric conversions (12 pts).
11. Slide 10 — Vocabulary flip cards: convert, conversion table, customary units, metric system, weight, mass, capacity, elapsed time.
12. Slide 11 — Vocabulary check (6 pts).
13. Slide 12 — Teach: hours and minutes (an hour has 60 minutes, not 100), elapsed time on a number line (6:40 to 9:15 = 2 hours 35 minutes), regrouping feet and inches (6 ft 15 in = 7 ft 3 in), and a four-step multi-step plan.
14. Slide 13 — Practice: time and mixed units (6 pts).
15. Slide 14 — Real World: the dockworker's clock, a two-step time problem (4 pts).
16. Slide 15 — Real World: weighing the fruit, a multi-step weight problem with tons and pounds (4 pts).
17. Slide 16 — Think It Through: "which explanation is correct?" and "find the mistake" (10 pts). Ask the student to say why each wrong choice is wrong.

18. Slide 17 — Fix the Sentence (8 pts): four sentences on quotation marks, its vs. it's, commas in a series, and subject-verb agreement. The answer must match exactly (straight or curly quotes both work).
19. Slide 18 — Match-Up Round (10 pts): 5 equal-measurement matches, 2 pts each on the first try.
20. Slide 19 — Speed Trail (bonus XP only).
21. Slide 20 — Looking Back reflection (bonus XP only).
22. Slide 21 — Boss Battle (8 pts): finishing time for the crane (minutes to hours and minutes, then elapsed time) and a two-step conversion problem.
23. Slide 22 — Pencil Break (Post): student completes the Post-Lesson Challenge sheet off screen.
24. Slide 23 — Results: score out of 100 sent to KingCretot Experience; 85+ shows the PortMiami congratulations with a big confetti burst; below 85 shows encouragement, and Play Again gives the next question set.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: MULTIPLYING FRACTIONS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: A whole number × a fraction (Week 13): type the product. 9 × 2/3 =	**6 (also accepted: 18/3, 6/1)**	9 × 2/3 = 18/3, and 18 ÷ 3 = 6.
r2_s0	Set 1: A fraction × a fraction (Week 13): 2/3 × 4/5 = ?	**8/15**	Multiply the numerators (2 × 4 = 8) and multiply the denominators (3 × 5 = 15). The product is 8/15.
r3_s0	Set 1: Does it make sense? (Week 13): which product is LESS than 10?	**10 × 3/4**	10 × 3/4 = 30/4 = 7 1/2, which is less than 10. That makes sense: 3/4 is less than 1 whole, so the product is less than 10. 1 1/4 and 1 1/2 are more than 1 whole, so those products are greater than 10, and 10 × 1 = 10.
r4_s0	Set 1: Fraction of a measurement: a mooring rope is 8 feet long. Andre coils 3/4 of it. How many feet does he coil? Feet:	**6**	3/4 of 8 means 3/4 × 8 = 24/4 = 6 feet.
r1_s1	Set 2: A whole number × a fraction (Week 13): type the product. 10 × 3/5 =	**6 (also accepted: 30/5, 6/1)**	10 × 3/5 = 30/5, and 30 ÷ 5 = 6.
r2_s1	Set 2: A fraction × a fraction (Week 13): 3/4 × 2/5 = ?	**3/10**	Multiply the numerators (3 × 2 = 6) and the denominators (4 × 5 = 20) to get 6/20. Simplify by dividing both by 2: 3/10.
r3_s1	Set 2: Does it make sense? (Week 13): which product is GREATER than 12?	**12 × 1 1/3**	12 × 1 1/3 = 12 × 4/3 = 48/3 = 16, which is greater than 12. That makes sense: 1 1/3 is more than 1 whole. 2/3, 1/2, and 5/6 are less than 1, so their products are less than 12.
r4_s1	Set 2: Fraction of a measurement: a dock line is 20 yards long. 2/5 of it is wet. How many yards are wet? Yards:	**8**	2/5 of 20 means 2/5 × 20 = 40/5 = 8 yards.
r1_s2	Set 3: A whole number × a fraction (Week 13): type the product. 12 × 5/6 =	**10 (also accepted: 60/6, 10/1)**	12 × 5/6 = 60/6, and 60 ÷ 6 = 10.

ID	Question	Answer	Why (explain to student)
r2_s2	Set 3: A fraction × a fraction (Week 13): 5/6 × 3/4 = ?	**5/8**	Multiply the numerators (5 × 3 = 15) and the denominators (6 × 4 = 24) to get 15/24. Simplify by dividing both by 3: 5/8.
r3_s2	Set 3: Does it make sense? (Week 13): which statement about 16 × 7/8 is true?	**It is less than 16**	16 × 7/8 = 112/8 = 14, which is less than 16. That makes sense: 7/8 is less than 1 whole, so the product is less than 16.
r4_s2	Set 3: Fraction of a measurement: a cargo strap is 12 feet long. Andre uses 3/4 of it. How many feet does he use? Feet:	**9**	3/4 of 12 means 3/4 × 12 = 36/4 = 9 feet.

YOUR TURN: INCHES, FEET, AND YARDS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: A 40-foot shipping container is 8 feet wide. How many inches wide is it? Inches:	**96**	1 foot = 12 inches. Feet are bigger than inches, so multiply: 8 × 12 = 96 inches.
l2_s0	Set 1: A dock line is 18 feet long. How many yards long is it?	**6 yards**	3 feet = 1 yard. Feet are smaller than yards, so divide: 18 ÷ 3 = 6 yards. (54 is what you get if you multiply by mistake.)
l3_s0	Set 1: Use the conversion table. How many inches are in 7 feet? [table: Feet = Inches; 1 = 12; 2 = 24; 5 = 60; 7 = ?] Inches in 7 feet:	**84**	Each foot adds 12 inches, so multiply by 12: 7 × 12 = 84 inches.
l1_s1	Set 2: A stack of crates on the dock is 6 feet tall. How many inches tall is it? Inches:	**72**	1 foot = 12 inches. Feet are bigger than inches, so multiply: 6 × 12 = 72 inches.
l2_s1	Set 2: The walkway from the dock to the ship is 27 feet long. How many yards long is it?	**9 yards**	3 feet = 1 yard. Feet are smaller than yards, so divide: 27 ÷ 3 = 9 yards. (81 is what you get if you multiply by mistake.)
l3_s1	Set 2: Use the conversion table. How many inches are in 9 feet? [table: Feet = Inches; 1 = 12; 3 = 36; 4 = 48; 9 = ?] Inches in 9 feet:	**108**	Each foot adds 12 inches, so multiply by 12: 9 × 12 = 108 inches.
l1_s2	Set 3: A pallet of pineapples is 5 feet tall. How many inches tall is it? Inches:	**60**	1 foot = 12 inches. Feet are bigger than inches, so multiply: 5 × 12 = 60 inches.
l2_s2	Set 3: Aunt Guerline lowers the crane hook 24 feet. How many yards is that?	**8 yards**	3 feet = 1 yard. Feet are smaller than yards, so divide: 24 ÷ 3 = 8 yards. (72 is what you get if you multiply by mistake.)
l3_s2	Set 3: Use the conversion table. How many feet are in 10 yards? [table: Yards = Feet; 1 = 3; 2 = 6; 6 = 18; 10 = ?] Feet in 10 yards:	**30**	Each yard adds 3 feet, so multiply by 3: 10 × 3 = 30 feet.

YOUR TURN: WEIGHT AND CAPACITY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: A box of bananas weighs 40 pounds. How many ounces does it weigh? Ounces:	**640**	1 pound = 16 ounces. Pounds are bigger than ounces, so multiply: 40 × 16 = 640 ounces.

ID	Question	Answer	Why (explain to student)
k2_s0	Set 1: An empty container on the port scale weighs 4 tons. How many pounds is that?	**8,000 pounds**	1 ton = 2,000 pounds. Tons are bigger than pounds, so multiply: 4 × 2,000 = 8,000 pounds.
k3_s0	Set 1: The dock crew drinks 24 quarts of water in one day. How many gallons is that? Gallons:	**6**	4 quarts = 1 gallon. Quarts are smaller than gallons, so divide: 24 ÷ 4 = 6 gallons.
k1_s1	Set 2: A box of pineapples weighs 20 pounds. How many ounces does it weigh? Ounces:	**320**	1 pound = 16 ounces. Pounds are bigger than ounces, so multiply: 20 × 16 = 320 ounces.
k2_s1	Set 2: A loaded 20-foot container of shoes weighs 6 tons. How many pounds is that?	**12,000 pounds**	1 ton = 2,000 pounds. Tons are bigger than pounds, so multiply: 6 × 2,000 = 12,000 pounds.
k3_s1	Set 2: Mr. Alvarez pours 16 pints of juice for the crew's lunch. How many quarts is that? Quarts:	**8**	2 pints = 1 quart. Pints are smaller than quarts, so divide: 16 ÷ 2 = 8 quarts.
k1_s2	Set 3: A box of avocados weighs 25 pounds. How many ounces does it weigh? Ounces:	**400**	1 pound = 16 ounces. Pounds are bigger than ounces, so multiply: 25 × 16 = 400 ounces.
k2_s2	Set 3: A forklift on the dock weighs 5 tons. How many pounds is that?	**10,000 pounds**	1 ton = 2,000 pounds. Tons are bigger than pounds, so multiply: 5 × 2,000 = 10,000 pounds.
k3_s2	Set 3: Mr. Alvarez pours 20 cups of water into a cooler. How many pints is that? Pints:	**10**	2 cups = 1 pint. Cups are smaller than pints, so divide: 20 ÷ 2 = 10 pints.

YOUR TURN: METRIC CONVERSIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: A rubber bumper on the side of the dock is 4 meters long. How many centimeters long is it? Centimeters:	**400**	1 meter = 100 centimeters. Meters are bigger, so multiply: 4 × 100 = 400 centimeters.
e2_s0	Set 1: Mr. Alvarez fills a 3-liter water jug before his shift. How many milliliters does it hold?	**3,000 milliliters**	1 liter = 1,000 milliliters. Liters are bigger, so multiply: 3 × 1,000 = 3,000 milliliters.
e3_s0	Set 1: A bag of rice in the ship's kitchen has a mass of 5,000 grams. What is its mass in kilograms? Kilograms:	**5**	1,000 grams = 1 kilogram. Grams are smaller, so divide: 5,000 ÷ 1,000 = 5 kilograms.
e1_s1	Set 2: A tugboat guides a ship the last 6 kilometers to the dock. How many meters is that? Meters:	**6,000**	1 kilometer = 1,000 meters. Kilometers are bigger, so multiply: 6 × 1,000 = 6,000 meters.
e2_s1	Set 2: A crate of mangoes has a mass of 7 kilograms. What is its mass in grams?	**7,000 grams**	1 kilogram = 1,000 grams. Kilograms are bigger, so multiply: 7 × 1,000 = 7,000 grams.
e3_s1	Set 2: Ms. Raman's water jug holds 4,000 milliliters. How many liters is that? Liters:	**4**	1,000 milliliters = 1 liter. Milliliters are smaller, so divide: 4,000 ÷ 1,000 = 4 liters.
e1_s2	Set 3: The shade cover over the crew's break area is 9 meters long. How many centimeters long is it? Centimeters:	**900**	1 meter = 100 centimeters. Meters are bigger, so multiply: 9 × 100 = 900 centimeters.

ID	Question	Answer	Why (explain to student)
e2_s2	Set 3: The soup pot in the ship's kitchen holds 5 liters. How many milliliters does it hold?	**5,000 milliliters**	1 liter = 1,000 milliliters. Liters are bigger, so multiply: 5 × 1,000 = 5,000 milliliters.
e3_s2	Set 3: A box of limes has a mass of 8,000 grams. What is its mass in kilograms? Kilograms:	**8**	1,000 grams = 1 kilogram. Grams are smaller, so divide: 8,000 ÷ 1,000 = 8 kilograms.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: The amount a container can hold, measured in cups, quarts, gallons, or liters, is called ___.	**Capacity**	Capacity is how much a container can hold. Cups, pints, quarts, gallons, milliliters, and liters are units of capacity.
v2_s0	Set 1: Type your answer: changing a measurement from one unit to another, like feet to inches, is called a ___.	**conversion (also accepted: unit conversion, conversions)**	Changing 3 feet into 36 inches is a conversion. The amount stays the same; only the unit changes.
v1_s1	Set 2: The amount of matter in an object, measured in grams and kilograms, is called ___.	**Mass**	Mass is the amount of matter in an object. Grams and kilograms are metric units of mass.
v2_s1	Set 2: Type your answer: meters, kilograms, and liters are all units in the ___ system.	**metric (also accepted: metric system)**	Meters, kilograms, and liters belong to the metric system, which is based on powers of 10.
v1_s2	Set 3: The amount of time that passes from the start of an event to the end is called ___.	**Elapsed time**	Elapsed time is how much time passes from start to finish, like from 6:40 a.m. to 9:15 a.m.
v2_s2	Set 3: Type your answer: feet, pounds, and gallons are ___ units, the system used most in the United States.	**customary (also accepted: U.S. customary, US customary, United States customary)**	Inches, feet, yards, ounces, pounds, tons, cups, pints, quarts, and gallons are customary units.

YOUR TURN: TIME AND MIXED UNITS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: How many minutes are in 2 hours 25 minutes? Minutes:	**145**	1 hour = 60 minutes, so 2 hours = 2 × 60 = 120 minutes. Then add the extra 25: 120 + 25 = 145 minutes.
t2_s0	Set 1: Andre joins two ropes. One is 5 ft 9 in long and the other is 3 ft 6 in long. How long are they together?	**9 ft 3 in**	Add feet and inches: 8 ft 15 in. 15 inches is more than a foot, so regroup 12 inches as 1 foot: 8 ft 15 in = 9 ft 3 in.
t1_s1	Set 2: How many minutes are in 3 hours 10 minutes? Minutes:	**190**	1 hour = 60 minutes, so 3 hours = 3 × 60 = 180 minutes. Then add the extra 10: 180 + 10 = 190 minutes.
t2_s1	Set 2: A crane works from 6:50 a.m. to 9:20 a.m. How much time is that?	**2 hours 30 minutes**	Count up: 6:50 to 7:00 is 10 minutes, 7:00 to 9:00 is 2 hours, and 9:00 to 9:20 is 20 minutes. 2 hours + 10 minutes + 20 minutes = 2 hours 30 minutes.

ID	Question	Answer	Why (explain to student)
t1_s2	Set 3: How many minutes are in 1 hour 50 minutes? Minutes:	**110**	1 hour = 60 minutes. Add the extra 50: 60 + 50 = 110 minutes.
t2_s2	Set 3: Andre joins two ropes. One is 4 ft 11 in long and the other is 3 ft 4 in long. How long are they together?	**8 ft 3 in**	Add feet and inches: 7 ft 15 in. 15 inches is more than a foot, so regroup 12 inches as 1 foot: 7 ft 15 in = 8 ft 3 in.

REAL WORLD: THE DOCKWORKER'S CLOCK (4 POINTS)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Aunt Guerline starts her crane shift at 6:00 a.m. She works 4 hours, takes a 30-minute lunch break, and then works 3 hours 45 minutes more. Two steps: change each part to minutes, then add. How many minutes long is her whole shift, including lunch? Minutes:	**495**	4 hours = 240 minutes and 3 hours 45 minutes = 180 + 45 = 225 minutes. Add the lunch break: 240 + 30 + 225 = 495 minutes.
w1_s1	Set 2: Mr. Alvarez checks the temperature inside each refrigerated container. Each check takes 6 minutes, and he has 25 containers to check. Two steps: how long will all 25 checks take, in hours and minutes?	**2 hours 30 minutes**	25 × 6 = 150 minutes. 150 minutes = 120 minutes + 30 minutes = 2 hours 30 minutes. (1 hour 50 minutes is a common mistake: there are 60 minutes in an hour, not 100.)
w1_s2	Set 3: Aunt Guerline's crane moves one container every 2 minutes. Her crew works a 3-hour stretch without stopping. Two steps: change the hours to minutes, then find how many containers the crane moves. Containers:	**90**	3 hours = 3 × 60 = 180 minutes. One container every 2 minutes means 180 ÷ 2 = 90 containers.

REAL WORLD: WEIGHING THE FRUIT (4 POINTS)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: A refrigerated container holds 20 pallets of bananas. Each pallet has 48 boxes, and each box weighs 40 pounds. Three steps: how much do all the bananas weigh, in tons and pounds?	**19 tons 400 pounds**	Boxes: 20 × 48 = 960. Pounds: 960 × 40 = 38,400 pounds. Tons: 38,400 ÷ 2,000 = 19 with 400 left over, so 19 tons 400 pounds.
w2_s1	Set 2: A refrigerated container holds 20 pallets of avocados. Each pallet has 60 boxes, and each box weighs 25 pounds. Three steps: how many tons do all the avocados weigh?	**15 tons**	Boxes: 20 × 60 = 1,200. Pounds: 1,200 × 25 = 30,000 pounds. Tons: 30,000 ÷ 2,000 = 15 tons.
w2_s2	Set 3: A truck leaves the port with 2 tons of limes. The limes are packed in boxes that weigh 40 pounds each. Two steps: change tons to pounds, then find how many boxes of limes are on the truck. Boxes:	**100**	2 tons = 2 × 2,000 = 4,000 pounds. Each box is 40 pounds: 4,000 ÷ 40 = 100 boxes.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? How many inches are in 2 yards?	**72 inches, because 2 yards = 6 feet and 6 × 12 = 72**	1 yard = 3 feet, so 2 yards = 6 feet. Each foot is 12 inches: 6 × 12 = 72 inches. (You can also do 2 × 36 = 72.) 36 inches is only 1 yard.

ID	Question	Answer	Why (explain to student)
x2_s0	Set 1: Find the mistake: Jamal says 5 kilograms = 500 grams. What did he do wrong?	**He used 100 instead of 1,000. 1 kilogram = 1,000 grams, so 5 kilograms = 5,000 grams**	1 kilogram = 1,000 grams, not 100. Kilograms are bigger, so multiply by 1,000: 5 × 1,000 = 5,000 grams.
x1_s1	Set 2: Which explanation is correct? Which is longer, 4 feet or 50 inches?	**50 inches, because 4 feet = 48 inches and 50 is greater than 48**	Change to the same unit first: 4 feet = 4 × 12 = 48 inches. 50 inches is greater than 48 inches, so 50 inches is longer.
x2_s1	Set 2: Find the mistake: Lucía says 48 ounces = 768 pounds, because 48 × 16 = 768. What did she do wrong?	**Ounces are smaller than pounds, so she should divide: 48 ÷ 16 = 3 pounds**	Going from a small unit (ounces) to a big unit (pounds) means you will have FEWER units, so divide: 48 ÷ 16 = 3 pounds.
x1_s2	Set 3: Which explanation is correct? A crane works from 7:15 a.m. to 9:00 a.m. How long is that?	**1 hour 45 minutes, because 7:15 to 8:15 is 1 hour and 8:15 to 9:00 is 45 minutes**	Count up: 7:15 to 8:15 is 1 hour, and 8:15 to 9:00 is 45 minutes, so the elapsed time is 1 hour 45 minutes. Subtracting 715 from 900 does not work because an hour has 60 minutes, not 100.
x2_s2	Set 3: Find the mistake: Mei says 2 ft 9 in + 1 ft 5 in = 3 ft 14 in, and that is her final answer. What should she fix?	**14 inches is more than a foot. Regroup 12 inches as 1 foot to get 4 ft 2 in**	1 foot = 12 inches, so 14 inches = 1 foot 2 inches. 3 ft 14 in = 4 ft 2 in.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix the sentence: andre asked, how many inches are in a yard	**Andre asked, "How many inches are in a yard?"**	Capitalize the first word and the name Andre, put quotation marks around his exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix the sentence: The crane lifted it's last container at noon.	**The crane lifted its last container at noon.**	Its (no apostrophe) shows ownership: the last container belongs to the crane. It's always means it is.
s3_s0	Set 1: Fix the sentence: The ship carried bananas pineapples and mangoes.	**The ship carried bananas, pineapples, and mangoes. OR The ship carried bananas, pineapples and mangoes.**	Use commas to separate the items in a series: bananas, pineapples, and mangoes.
s4_s0	Set 1: Fix the sentence: the containers on the dock weighs 8,000 pounds each	**The containers on the dock weigh 8,000 pounds each. OR The containers on the dock weighed 8,000 pounds each.**	Capitalize the first word. The subject is containers (plural), so use weigh, not weighs. End with a period.
s1_s1	Set 2: Fix the sentence: aunt guerline said, every crane gets a safety check	**Aunt Guerline said, "Every crane gets a safety check."**	Capitalize Aunt Guerline (a title used with a name), put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
s2_s1	Set 2: Fix the sentence: Its about 60 minutes until the ship docks.	**It's about 60 minutes until the ship docks.**	It's means it is: it is about 60 minutes. Use the apostrophe.
s3_s1	Set 2: Fix the sentence: A gallon is equal to 4 quarts 8 pints or 16 cups.	**A gallon is equal to 4 quarts, 8 pints, or 16 cups. OR A gallon is equal to 4 quarts, 8 pints or 16 cups.**	Use commas to separate the items in a series: 4 quarts, 8 pints, or 16 cups.
s4_s1	Set 2: Fix the sentence: mr. alvarez check the refrigerated containers every hour	**Mr. Alvarez checks the refrigerated containers every hour. OR Mr. Alvarez checked the refrigerated containers every hour.**	Capitalize the first word and the name Mr. Alvarez. One person (Mr. Alvarez) needs the verb checks, not check. End with a period.
s1_s2	Set 3: Fix the sentence: andre said, a kilometer is 1,000 meters	**Andre said, "A kilometer is 1,000 meters."**	Capitalize Andre, put quotation marks around his exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s2	Set 3: Fix the sentence: Each container has it's own number.	**Each container has its own number.**	Its (no apostrophe) shows ownership: the number belongs to the container. It's means it is.
s3_s2	Set 3: Fix the sentence: The truck drivers came from Miami Doral and Hialeah.	**The truck drivers came from Miami, Doral, and Hialeah. OR The truck drivers came from Miami, Doral and Hialeah.**	Use commas to separate the items in a series: Miami, Doral, and Hialeah.
s4_s2	Set 3: Fix the sentence: the dockworkers was ready at 6:00 in the morning	**The dockworkers were ready at 6:00 in the morning. OR The dockworker was ready at 6:00 in the morning.**	Capitalize the first word. The subject is dockworkers (plural), so use were, not was. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH, 10 TOTAL)

ID	Question	Answer	Why (explain to student)
match1_s0	Set 1: 1 yard	**3 feet**	1 yard = 3 feet.
match2_s0	Set 1: 1 foot	**12 inches**	1 foot = 12 inches.
match3_s0	Set 1: 1 pound	**16 ounces**	1 pound = 16 ounces.
match4_s0	Set 1: 1 gallon	**4 quarts**	1 gallon = 4 quarts.
match5_s0	Set 1: 1 kilometer	**1,000 meters**	1 kilometer = 1,000 meters.
match1_s1	Set 2: 1 ton	**2,000 pounds**	1 ton = 2,000 pounds.
match2_s1	Set 2: 1 quart	**2 pints**	1 quart = 2 pints.
match3_s1	Set 2: 1 meter	**100 centimeters**	1 meter = 100 centimeters.
match4_s1	Set 2: 1 liter	**1,000 milliliters**	1 liter = 1,000 milliliters.

ID	Question	Answer	Why (explain to student)
match5_s1	Set 2: 1 hour	**60 minutes**	1 hour = 60 minutes.
match1_s2	Set 3: 2 yards	**6 feet**	2 × 3 = 6 feet.
match2_s2	Set 3: 3 feet	**36 inches**	3 × 12 = 36 inches.
match3_s2	Set 3: 2 pounds	**32 ounces**	2 × 16 = 32 ounces.
match4_s2	Set 3: 4 gallons	**16 quarts**	4 × 4 = 16 quarts.
match5_s2	Set 3: 3 kilograms	**3,000 grams**	3 × 1,000 = 3,000 grams.

BOSS BATTLE: THE BIG UNLOADING (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: The cargo ship Caribe Star is tied up at the dock. Starting at 8:00 a.m., Aunt Guerline's crane must unload 75 containers without stopping, and each lift takes 2 minutes. Inside one refrigerated container, a pallet of pineapples has a mass of 600 kilograms. Two steps: at what time does the crane finish unloading?	**10:30 a.m.**	75 × 2 = 150 minutes. 150 minutes = 2 hours 30 minutes. 8:00 a.m. + 2 hours 30 minutes = 10:30 a.m. (9:50 comes from thinking 150 minutes is 1 hour 50 minutes.)
b2_s0	Set 1: The cargo ship Caribe Star is tied up at the dock. Starting at 8:00 a.m., Aunt Guerline's crane must unload 75 containers without stopping, and each lift takes 2 minutes. Inside one refrigerated container, a pallet of pineapples has a mass of 600 kilograms. Two steps: each box of pineapples has a mass of 12,000 grams. How many boxes are on the 600-kilogram pallet? Boxes:	**50**	12,000 grams = 12,000 ÷ 1,000 = 12 kilograms per box. Then 600 ÷ 12 = 50 boxes.
b1_s1	Set 2: The cargo ship Bay Runner docks at 6:45 a.m. Starting at 7:00 a.m., Aunt Guerline's crane unloads 50 containers without stopping, and each lift takes 3 minutes. Mr. Alvarez fills 3 water coolers for the crew, and each cooler holds 4 gallons. Two steps: at what time does the crane finish unloading?	**9:30 a.m.**	50 × 3 = 150 minutes. 150 minutes = 2 hours 30 minutes. 7:00 a.m. + 2 hours 30 minutes = 9:30 a.m. (8:50 comes from thinking 150 minutes is 1 hour 50 minutes.)
b2_s1	Set 2: The cargo ship Bay Runner docks at 6:45 a.m. Starting at 7:00 a.m., Aunt Guerline's crane unloads 50 containers without stopping, and each lift takes 3 minutes. Mr. Alvarez fills 3 water coolers for the crew, and each cooler holds 4 gallons. Two steps: how many quarts of water are in all 3 coolers? Quarts:	**48**	3 coolers × 4 gallons = 12 gallons. 1 gallon = 4 quarts, so 12 × 4 = 48 quarts.
b1_s2	Set 3: The cargo ship Coral Wind docks at 9:00 a.m. Starting at 9:15 a.m., Aunt Guerline's crane unloads 40 containers without stopping, and each lift takes 4 minutes. Andre has a 5-yard rope to cut into pieces for tying down tarps. Two steps: at what time does the crane finish unloading?	**11:55 a.m.**	40 × 4 = 160 minutes. 160 minutes = 2 hours 40 minutes. 9:15 a.m. + 2 hours 40 minutes = 11:55 a.m. (11:15 comes from thinking 160 minutes is 1 hour 60 minutes, which is really only 2 hours.)

ID	Question	Answer	Why (explain to student)
b2_s2	Set 3: The cargo ship Coral Wind docks at 9:00 a.m. Starting at 9:15 a.m., Aunt Guerline's crane unloads 40 containers without stopping, and each lift takes 4 minutes. Andre has a 5-yard rope to cut into pieces for tying down tarps. Two steps: Andre cuts the 5-yard rope into pieces that are each 18 inches long. How many pieces does he get? Pieces:	**10**	5 yards = 5 × 36 = 180 inches. Then 180 ÷ 18 = 10 pieces.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Fill in: ft, yd, lb, hour	1 foot = 12 inches, 1 yard = 3 feet, 1 pound = 16 ounces, 1 hour = 60 minutes.
2	4 feet to inches	48 inches (4 × 12 = 48).
3	Unit for the water in a cooler	gallons
4	Fill in: m, kg, L	1 meter = 100 centimeters, 1 kilogram = 1,000 grams, 1 liter = 1,000 milliliters.
5	Match the words	capacity: how much a container can hold; mass: the amount of matter in an object, measured in grams or kilograms; elapsed time: the time that passes from start to finish.
6	2/3 × 12	8 (2/3 × 12 = 24/3 = 8).

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	40 feet to inches, and to yards and feet	480 inches (40 × 12). 40 ÷ 3 = 13 R1, so 13 yards 1 foot.
2	Heaviest	3 tons (= 6,000 pounds). 90,000 ounces = 5,625 pounds; 2 tons 900 pounds = 4,900 pounds.
3	6:45 a.m. to 3:15 p.m. minus 30-minute lunch	8 hours. 6:45 a.m. to 3:15 p.m. is 8 hours 30 minutes; minus the 30-minute lunch leaves 8 hours.
4	3 gallons in cups; 40 cups used	48 cups (3 × 16 = 48). 48 − 40 = 8 cups left over.
5	Writing: kilograms to grams	Jamal is wrong. Model: "Jamal is not right. Kilograms are bigger than grams, so you multiply by 1,000 to get more of the smaller unit. For example, 3 kilograms = 3 × 1,000 = 3,000 grams." Look for complete sentences, the word multiply, and a correct numerical example.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Perimeter of a 120 yd by 50 yd yard	340 yards (120 + 50 + 120 + 50). 340 × 3 = 1,020 feet. 1,020 × 12 = 12,240 inches.